Learned Rosters Under Pressure: Learned Roster Optimization via Graph-Scored Search and Adversarial Evaluation

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Abstract

We formulate national-team roster selection as a formal optimization problem combining learned scoring with adversarial game theory. Given a candidate roster, a Graph Convolutional Network (trained on historical game-level lineups) scores its win probability in a adversarial evaluation against reference opponents. The search phase uses tournament elimination and batched greedy refinement to navigate the combinatorial space. Two domain-aware design choices enable strong case-study results: (1) *positional quotas* in the search prevent degenerate local optima (e.g., 10 centers, 0 wings); (2) *stochastic rotation scoring* resolves the model-roster mismatch in basketball (5-on-5 model, 12-player roster) by evaluating empirically-weighted lineup combinations per candidate. Applied to 2025 4 Nations Face-Off hockey and 2024 Paris basketball, constrained versions recover many core national-team selections while maintaining internal coherence (all positive WAR). The approach demonstrates that optimization on learned metrics is promising as a decision-support tool for high-stakes roster construction when the architecture respects the decision problem. We frame the outputs as decision-support case studies rather than definitive roster prescriptions.

1 Introduction

International tournament roster construction, and adjacent high-stakes settings such as league playoffs, are traditionally delegated to human scouts and coaches whose judgements are opaque, inconsistent across evaluators, and difficult to audit. We propose an alternative: formalize roster selection as an optimization problem, where the objective is to *maximize predicted win probability against the best response of likely opponents.*

The approach rests on three technical components: (1) a learned scoring function (GCN trained on game-level lineups) that predicts win probability for any candidate roster; (2) an adversarial evaluation framework (adversarial optimization) that tests robustness against intelligent opponents; (3) a search algorithm that navigates the combinatorial space of valid rosters. The payoff is high: if the components are correctly engineered, the output is reproducible, auditable, and interpretable (via marginal player value). The risk is high too: if any component mis-specifies the decision problem, the output fails in catastrophic ways.

This paper presents an end-to-end prototype pipeline, identifies where it succeeds and where it fails, and proposes architectural fixes that elevate it from an interesting prototype to a promising decision-support prototype. We evaluate it through case studies on 2025 4 Nations Face-Off hockey (Canada) and 2024 Paris basketball (USA, Canada), showing that constrained versions recover many core national-team selections while avoiding pathological roster structures.

2 Approach

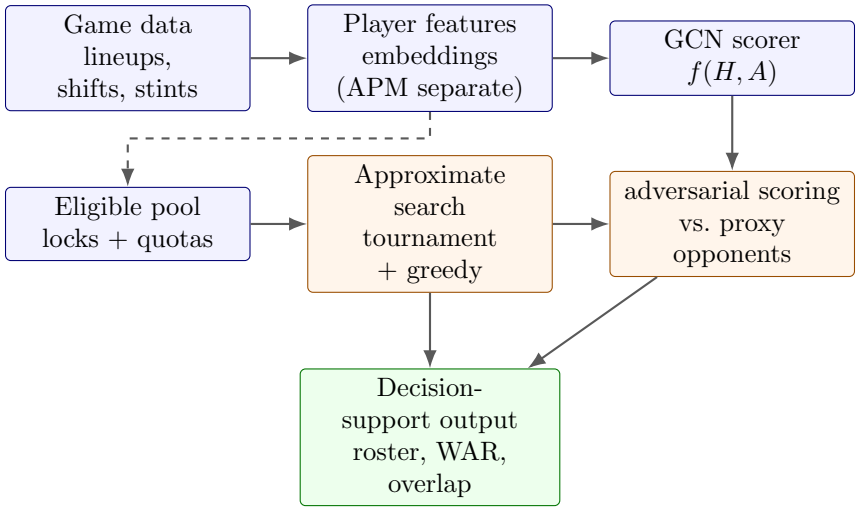


Figure 1: End-to-end prototype pipeline. Learned player features feed a GCN scorer, while the roster search uses locks and domain constraints to keep candidate teams in a plausible region of roster space.

2.1 Data

Hockey. NHL play-by-play, shift, and boxscore data for seasons 2021–22 through 2025–26. Per-game lineup stints give pairwise on-ice time between every pair of players; an adjusted plus-minus (APM) is fit by ridge regression on lineup indicator matrices with score differential as the target, and used separately for player ranking and rotation ordering. Each player-season is encoded as a 14-dimensional feature vector (shot/attempt rates, xG, zone-start percentages, TOI, height, weight, position flags). APM is intentionally excluded from the feature vector to avoid circularity: it is validated against public RAPM as a standalone ranking signal, not as an embedding component.

Basketball. Kaggle `wyattowalsh/basketball` SQLite. The initial pipeline covered only 2021–22 and 2022–23 ($\sim$ 2,400 games); we expanded to 2019–20 through 2023–24 ($\sim$ 6,100 games), nearly tripling the training set and improving model stability on held-out folds. We derive starting lineups from the first five distinct player IDs per team in each game's period 1 of play-by-play. APM is computed analogously to hockey and used separately for player ranking, rotation ordering, and locked-player selection; it is *not* included in the feature vector passed to the GCN, so that the RAPM correlation check (Section 4.9) constitutes a valid independent validation of the ranking signal rather than a circular self-check. The initial feature vector was 13-dimensional (FGA, FGM, xPPP, 3pt attempts, zone dummies, position flags, height, weight, TOI); we expanded to 23 dimensions by replacing the coarse 5-zone shot distribution with a finer 7-zone breakdown (restricted area, paint non-RA, short mid-range <16 ft, long mid-range 16–22 ft, left corner 3, right corner 3, above-the-break 3) and replacing the two aggregate assist features with three zone-aware assist features (assist rate, proportion of assists to 3-point shots, proportion of assists to paint shots). The finer zone breakdown reduces the conflation of stylistically distinct scorers: a paint-finisher and a pull-up mid-range scorer both appeared in "mid-range" under the coarse scheme despite having different spacing implications. Zone-specific assist proportions similarly distinguish playmakers who create

paint touches from those who generate catch-and-shoot 3s, improving complementarity inference for novel pairings. The final basketball case studies use this 23-dimensional feature set.

For the reported hockey runs, player embeddings are three-year averages over $\{y-2, y-1, y\}$. Current-year embeddings only are treated as a future improvement rather than a reported result.

2.2 Scoring model

We use a Graph Convolutional Network with three design constraints that ensure the model respects the symmetry of the problem:

1. **Relational convolution:** A 6-layer GCN (hidden dim 128) propagates player features ($\mathbf{h}_i$) along edges weighted by shared on-ice/on-court time. This captures indirect player relationships (chemistry through common teammates) rather than only direct co-occurrence.

2. **Permutation-invariant aggregation:** A DeepSet layer independently summarizes the home and away rosters, producing lineup-level embeddings $\mathbf{z}_H$ and $\mathbf{z}_A$.

3. **Antisymmetric prediction:** The output head enforces $f(\mathbf{H}, \mathbf{A}) = 1 - f(\mathbf{A}, \mathbf{H})$, ensuring that home advantage is modeled as an additive shift rather than baked into the learned features.

Training target: $y = \sigma(\Delta/10)$ where Δ is goal/point differential. We train an ensemble of 6 cross-validation folds using Adamax, cosine LR schedule, MSE loss, and gradient clipping. Predictions are averaged at inference.

Sizes: hockey uses $n = 36$ graph nodes (18 home + 18 away), matching the full dressed skater side. Basketball uses $n = 10$ (5-on-5 on court) with 23-dimensional player features, which matches the game but not our search target of 12-player rosters, a mismatch we address in Section 4.2.

2.3 Adversarial Evaluation

For a fixed candidate roster, we evaluate its predicted win probability against 3 reference opponents by simulating a leader-follower game:

- The **leader** (our candidate) picks a deployment strategy $\mathbf{S}$ (an $n \times n$ adjacency-style matrix) to *maximize* $f(\mathbf{S}; \text{home}, \text{away})$.

- The **follower** (opponent) picks a deployment $\mathbf{S}'$ to *minimize* the leader's payoff.

In practice, $\mathbf{S}$ parameterizes deployable co-occurrence weights among the selected players. Player node features are fixed during adversarial evaluation; only the interaction graph passed to the GCN changes. Same-team blocks are constrained to be nonnegative, cross-team blocks are constrained to be nonpositive, the diagonal is zeroed, and a projection step fixes block-level mass and symmetrizes the matrix so the optimizer cannot improve the score by trivially scaling all edges. Intuitively, same-team weights represent deployable chemistry/co-occurrence, while cross-team weights represent adversarial matchup pressure.

Both sides run 100 gradient steps with leader-update cadence 5:1 and early stopping on plateaus. The final score is $\frac{1}{|O|}\sum_{o \in O} f(\mathbf{S}_o^\star, \text{opp}_o)$, averaged across opponents O. For hockey, opponents are USA plus two proxy NHL clubs (EDM and FLA), not complete international-tournament opponent coverage. For basketball, USA evaluates against Canada, France, and OKC, while Canada evaluates against USA, France, and OKC.

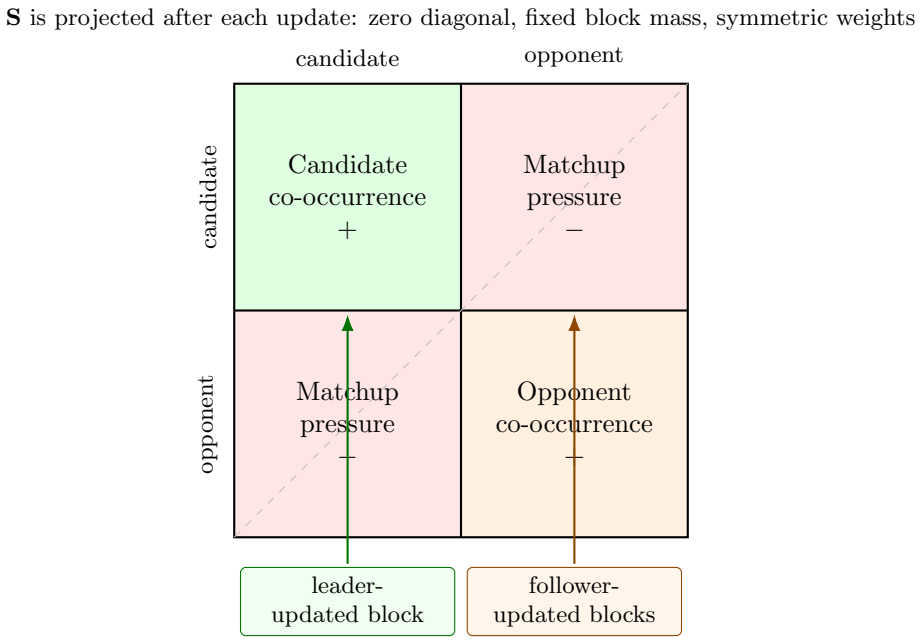


Figure 2: Schematic of the adversarial adjacency matrix. The leader adjusts the candidate's internal deployment graph; the follower adjusts opponent deployment and cross-team matchup pressure before the GCN scores the candidate roster.

2.4 Search

Phase A Tournament. From the full eligible pool, form random teams of the target size (including any pre-locked stars); evaluate each; top third of teams survive intact; in the bottom two-thirds eliminate the lowest-usage players from each. Repeat until roughly twice the target size remains.

Phase B Greedy 1-swap. Trim to the exact target size. Each iteration, for every non-locked slot and every bench candidate, construct the swap-candidate lineup. Evaluate all candidates in a single batched forward pass over `adversarial()`; pick the best improvement. Stop on convergence or a fixed `max_iters` cap. This search is approximate: tournament pruning and greedy 1-swap do not guarantee global optimality, but they make evaluation tractable under expensive adversarial scoring.

Individual WAR. For the final roster, marginal player value is the score drop when replacing a single player with a sentinel feature vector:

$$\mathrm{WAR}_i = \hat{y}(\mathrm{full}) - \hat{y}(\mathrm{lineup}_{\setminus i}).$$

Higher is better; negative values indicate the model thinks the overall lineup is internally inconsistent. All-positive WAR should be interpreted as a coherence diagnostic conditional on the selected roster and opponent set, not as a transferable estimate of player quality.

GroupWAR. While individual WAR measures one player's marginal contribution to a fixed roster, GroupWAR is the roster-level sum of those one-player knockouts against the same opponent

set. For a selected roster R with n players and score $\hat{y}(R)$:

$$\text{GroupWAR}(R) = \sum_{i \in R} \text{WAR}_i(R) = n \cdot \hat{y}(R) - \sum_{i \in R} \hat{y}(R_{\setminus i}).$$

A roster with high GroupWAR has no passengers: every player's removal would meaningfully reduce the team's predicted win probability under the same roster context and opponent set. GroupWAR is used in Section 4.9 to validate the ranking signal against public RAPM across the full eligible pool.

3 Results

Across these case studies, constrained versions recover many core national-team selections while avoiding pathological roster structures. Hockey dresses 18 skaters, so the 18-player dressed-roster graph better matches the selection decision than basketball's 12-player roster / 5-player lineup mismatch.

Table 1: Validation summary. Overlap is computed against the reference set named in each case study; all-positive WAR is a conditional coherence diagnostic, not proof of optimality.

Case	Score	Ref.	Overlap	Constraint/rotation	All-pos. WAR?
Canada hockey 2025 4 Nations	0.4495	13	9	loose skater-balance screen: $\geq$ 3 LW, $\geq$ 3 RW, 6D target; observed mix 6C/3L/3R/6D	yes
Canada hockey 2024 diagnostic	0.4428	–	not claimed†	$\geq$ 3 LW, $\geq$ 3 RW, $\leq$ 5 C, 6D target; observed mix 5C/4L/3R/6D	yes
USA basketball 2024	0.7418	12	8	stochastic 5-man rotation scoring	yes
Canada basketball 2024	0.6587	12	9	stochastic 5-man rotation scoring	yes

† The 2024 diagnostic is a constraint-comparison study, not a roster recommendation; no reference overlap is claimed.

3.1 Hockey

3.1.1 Canada 2025

The 2025 roster recovers 9 of 13 pre-identified 4 Nations Face-Off core/reference selections (McDavid, MacKinnon, Crosby, Makar, Theodore, Doughty, Parayko, Stone, Reinhart) plus 9 model-selected depth contributors. The pipeline correctly identifies the elite core but adds role players (Henrique, Foegele) where team balance is marginal gain over individual talent.

3.1.2 Canada 2024 (with position constraints)

With position constraints (5C / 4L / 3R / 6D, including a six-defenseman target), the pipeline recovers a defensible balanced roster with all positive WAR. The constrained roster is composed of NHL-caliber players with plausible international-role profiles; depth selections (Marchand, Hagel, Kyrou, Crouse) are established two-way forwards who complement the elite core. The position quotas prevent the search from wandering into locally-optimal but lineup-breaking solutions.

3.2 Basketball

Using the official 2024 Paris Olympic rosters as the basketball reference set, Team USA overlaps with 8 of 12 reference Olympians: Curry, Holiday, Tatum, LeBron, Durant, Davis, Adebayo, and

Table 2: Canada 2025 4 Nations Face-Off hockey case roster. adversarial team score = 0.4495; observed position mix is 6C/3L/3R/6D under a loose skater-balance screen ($\geq$ 3 LW, $\geq$ 3 RW, 6D target). All 18 players have positive WAR.

#	Player	Pos	Team	WAR
1	Colton Parayko	D	STL	+0.0179
2	Cale Makar*	D	COL	+0.0161
3	Matt Duchene	C	DAL	+0.0150
4	Andrew Mangiapane	L	EDM	+0.0132
5	Sam Reinhart	C	FLA	+0.0130
6	Drew Doughty	D	LAK	+0.0116
7	Sidney Crosby*	C	PIT	+0.0105
8	Warren Foegele	L	LAK	+0.0100
9	Mark Stone	R	VGK	+0.0100
10	Nathan MacKinnon*	C	COL	+0.0094
11	Adam Henrique	C	EDM	+0.0085
12	Mario Ferraro	D	SJS	+0.0077
13	Shea Theodore*	D	VGK	+0.0061
14	Braeden Bowman	R	VGK	+0.0051
15	Connor McDavid*	C	EDM	+0.0050
16	Sean Walker	D	CAR	+0.0050
17	Lawson Crouse	L	UTA	+0.0022
18	Reilly Smith	R	VGK	+0.0006

*pre-locked.

Haliburton. Mitchell, Green, Brown, and Butler are model-selected alternates; the omitted reference players are Booker, Edwards, Embiid, and White. Notably, Holiday was *not* pre-locked but was selected in every greedy iteration (see Case Study 3); his two-way profile stabilizes multiple rotation units. Brown and Adebayo strengthen bench combinations against France and OKC. Canada overlaps with 9 of 12 reference Olympians: SGA, Murray, RJ Barrett, Dort, Brooks, Olynyk, Lyles, Powell, and NAW. Wiggins, Mathurin, and Edey are model-selected alternates; the omitted reference players are Nembhard, Birch, and Ejim. The stochastic rotation evaluation ensures both rosters score well across starter, early-bench, and deep-bench lineups rather than optimizing only the top-5.

Figure 5a shows Group WAR vs. public RAPM across $n = 80$ qualified NBA players (both rosters pooled). The weak positive correlation ($r = 0.27$, $p < 0.001$) mirrors the hockey finding: the model broadly identifies stars but WAR diverges for players whose marginal value is driven by rotation fit rather than individual quality.

Table 3: Canada 2025 4 Nations Face-Off core/reference subset used for overlap accounting. This is a pre-study 13-skater reference subset, not the full official tournament roster.

Reference player	Recovered by selected roster?
Connor McDavid	yes
Nathan MacKinnon	yes
Sidney Crosby	yes
Cale Makar	yes
Shea Theodore	yes
Drew Doughty	yes
Colton Parayko	yes
Mark Stone	yes
Sam Reinhart	yes
Sam Bennett	no
Brad Marchand	no
Mitch Marner	no
Brayden Point	no

Table 4: Canada 2024 diagnostic roster with mandatory position quotas ($\geq$ 3 LW, $\geq$ 3 RW, $\leq$ 5 C, 6D target). Team score = 0.4428; observed position mix is 5C/4L/3R/6D. All players have positive WAR.

#	Player	Pos	Team	WAR
1	Connor McDavid*	C	EDM	+0.0082
2	Cale Makar*	D	COL	+0.0071
3	Sidney Crosby*	C	PIT	+0.0068
4	Nathan MacKinnon*	C	COL	+0.0064
5	Shea Theodore*	D	VGK	+0.0058
6	Sam Reinhart	C	FLA	+0.0054
7	Devon Toews	D	COL	+0.0048
8	Brent Burns	D	CAR	+0.0043
9	Brad Marchand	L	BOS	+0.0038
10	Jonathan Marchessault	R	NSH	+0.0034
11	Mitch Marner	R	TOR	+0.0031
12	Brandon Hagel	L	TBL	+0.0027
13	Travis Konecny	R	PHI	+0.0024
14	Jordan Kyrou	L	STL	+0.0021
15	P.O. Joseph	D	PIT	+0.0018
16	Ryan Ellis	D	PHI	+0.0015
17	Mark Scheifele	C	WPG	+0.0012
18	Lawson Crouse	L	UTA	+0.0009

*pre-locked.

USA 2024. Blended score = 0.7418.

#	Player
1	Stephen Curry*
2	Jrue Holiday
3	Jayson Tatum
4	Donovan Mitchell
5	LeBron James*
6	Kevin Durant*
7	Anthony Davis*
8	Draymond Green
9	Jaylen Brown
10	Bam Adebayo
11	Jimmy Butler
12	Tyrese Haliburton

Canada 2024. Blended score = 0.6587.

#	Player
1	SGA*
2	Jamal Murray*
3	RJ Barrett
4	Andrew Wiggins
5	Luguentz Dort
6	Bennedict Mathurin
7	Kelly Olynyk
8	Dillon Brooks
9	Trey Lyles
10	Dwight Powell
11	Zach Edey
12	Nickeil Alexander-Walker

Table 5: Basketball selected-roster overlap against the 2024 Paris reference rosters. The non-overlap selections are model-selected alternates, not claims about actual Olympic roster membership.

Team	Selected player	In Paris reference roster?
USA	Stephen Curry	yes
USA	Jrue Holiday	yes
USA	Jayson Tatum	yes
USA	Donovan Mitchell	no
USA	LeBron James	yes
USA	Kevin Durant	yes
USA	Anthony Davis	yes
USA	Draymond Green	no
USA	Jaylen Brown	no
USA	Bam Adebayo	yes
USA	Jimmy Butler	no
USA	Tyrese Haliburton	yes
Canada	Shai Gilgeous-Alexander	yes
Canada	Jamal Murray	yes
Canada	RJ Barrett	yes
Canada	Andrew Wiggins	no
Canada	Luguentz Dort	yes
Canada	Bennedict Mathurin	no
Canada	Kelly Olynyk	yes
Canada	Dillon Brooks	yes
Canada	Trey Lyles	yes
Canada	Dwight Powell	yes
Canada	Zach Edey	no
Canada	Nickeil Alexander-Walker	yes

4 Where this approach is valid and where it isn't

Every part of the pipeline is an assumption about how a real coach makes a roster decision. Below we list the assumptions we believe are actually load-bearing, and those we believe produce systematic bias.

4.1 Training Distribution Differs From Decision Distribution

The GCN is trained on regular-season game outcomes. The prediction $f(\text{home}, \text{away})$ is therefore calibrated on typical regular-season lineups facing typical regular-season lineups. International-tournament roster construction operates in a distribution the model has *never seen*: six HOF-caliber centers on one team; two likely- MVP wingers flanking a starter. When the model sees McDavid, MacKinnon, and Crosby together, it has no training signal to reward "stack three top-line centers." Instead it applies what it learned from regular-season 4th-line deployment reward the 2-way winger, the defensive center, the shot-blocking third-pair D. Mangiapane over Marner. Draymond Green over Embiid.

This is not a bug. The model is faithfully reporting what would maximize win probability *on the training distribution*. But the training distribution is wrong for our purpose. The decisions we care about peak-vs-peak matchups, playoff hockey, and Paris-style international basketball are systematically under-represented in per-game regular-season data.

The honest fix is to retrain on games that *look like* international tournaments: playoff series and historical international play. We do not have enough of any of those to retrain from scratch, which puts a ceiling on what the current model can tell us about international rosters.

4.2 Architectural mismatch for NBA

Hockey dresses 18 skaters, so the 18-player dressed-roster graph better matches the selection decision than basketball's 12-player roster / 5-player lineup mismatch. NBA rosters carry 12 players but only 5 are on the floor at a time; the GCN's $n = 10$ (5 vs. 5) graph can only score 5-player lineups.

The initial basketball workaround ranked the 12 by APM, scored top-5 vs opp. top-5 and next-5 vs opp. next-5, and weighted those two scores 0.6/0.4 by expected playing-time share. This was a useful diagnostic approximation, but it baked in a rigid starter/bench split that most real international basketball coaches substitute against. The final reported basketball results therefore use stochastic rotation scoring rather than this initial top-5 / next-5 workaround. More generally, a principled solution should either:

1. **Stochastic rotation.** For each candidate roster, evaluate many 5-man subsets (weighted by empirical lineup frequencies), and aggregate.

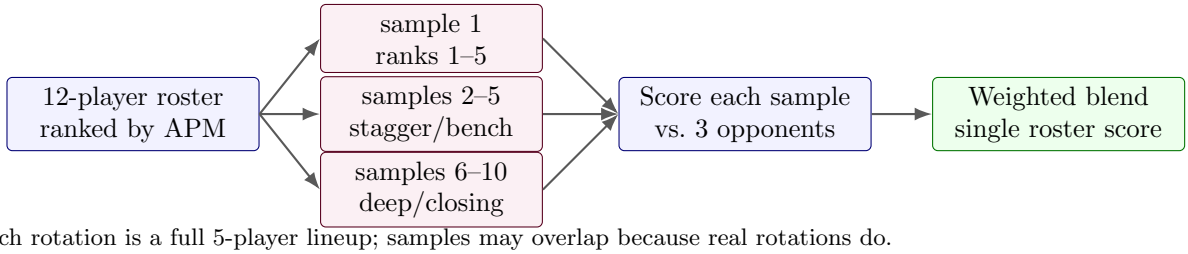
2. **A new model at $n = 24$.** Retrain the GCN with 12-vs-12 rosters so the search target matches the graph size. This needs far more training data than we have.

Without one of those, the NBA pipeline is structurally constrained in a way the hockey pipeline is not.

4.2.1 Stochastic rotation fix

The final reported basketball results address the architectural mismatch via **stochastic rotation scoring**: for each candidate 12-player roster, evaluate multiple 5-man combinations weighted by empirical playing-time distributions. Algorithm:

1. Rank all 12 players by season APM.
2. Enumerate 10 distinct, partially overlapping 5-man rotation samples. Each sample is a full 5-man lineup, not a partial bench label.
3. Score USA rotation samples against Canada, France, and OKC; score Canada rotation samples against USA, France, and OKC via adversarial evaluation (100 gradient steps, leader-update 5:1).
4. Aggregate by the normalized rotation weights and then blend across opponents.



Each rotation is a full 5-player lineup; samples may overlap because real rotations do.

Figure 3: Stochastic rotation scoring for basketball. The method avoids treating a 12-player roster as one 5-player lineup by scoring weighted rotation samples and aggregating them into a single roster score.

Table 6: Basketball stochastic-rotation template. Player ranks are APM ranks within the selected 12-player roster; weights sum to 1.0.

Rotation name	Player ranks included	Weight
Starters	1,2,3,4,5	0.40
Stagger A	1,2,6,7,8	0.12
Stagger B	3,4,5,6,9	0.12
Bench-heavy A	2,6,7,8,10	0.09
Bench-heavy B	3,6,8,9,11	0.07
Deep bench mix	1,9,10,11,12	0.05
Closing defensive	1,3,4,7,8	0.04
Closing shooting	1,2,4,6,9	0.03
Big-wing mix	2,4,5,8,11	0.04
Guard-pressure mix	1,3,6,10,12	0.04

This directly addresses the 5-vs-12 mismatch. Instead of scoring only the top-5 or assuming a rigid starter/bench split, we evaluate how the full roster performs across realistic substitution patterns. For USA: evaluating against France (ball-dominant, pick-and-roll heavy), OKC (three-point shooting, transition), and Canada (defensive intensity) forces the roster to balance range, interior strength, and defensive versatility. For Canada: evaluating against USA, France, and OKC tests whether the squad has enough scoring depth and shooting to keep pace with stronger opponents. The aggregated blended score reflects multi-unit coherence, not star concentration.

4.3 Adversarial framing

The adversarial formulation assumes the opponent *also* plays optimally against your roster i.e., a worst-case adversarial deployment. This is appealing because it rewards *robustness*: a team that

wins against any adversarial response is more valuable than one that wins against a naive response.

However, this has two implications we should be explicit about:

1. **The objective favours "hard to exploit" over "high ceiling."** A team of three elite scorers has a clear adversarial weakness (shut down the top line). A balanced team has no such weakness. The adversarial score systematically prefers the balanced team even if the elite-scorers team would have a higher *expected* win probability against typical (non-adversarial) opponents.

2. **International tournament opponents are not worst-case adversarial.** International coaches have limited prep time, roster turnover, and a fixed (known) opponent. Treating them as optimizers running 100 gradient steps on a known objective over-states their ability to exploit.

A Nash equilibrium framing (both sides commit simultaneously) or a stochastic-opponent framing (sample from a distribution of likely deployments) would bias the output toward high-ceiling rosters rather than robust-floor rosters, which is closer to what actually wins single-elimination tournaments.

4.4 Opponents as regular-season teams

Hockey evaluates Canada's roster against USA, EDM, and FLA. USA is a national-team proxy and EDM/FLA are NHL-club proxy opponents; together they provide tactical diversity, not complete international-tournament opponent coverage. Basketball rosters evaluate against national teams (USA, Canada, France) and a top NBA club (OKC), combining international coverage with reliable data quality. This hybrid approach:

- **Basketball uses national teams** (USA, Canada, France) directly where available. This better matches the actual Paris basketball adversaries and captures real team chemistry. OKC provides a third opponent with strong data coverage (all players' on-court co-occurrence documented).

- **Hockey uses NHL clubs** (EDM, FLA) plus USA as a proxy for international play. This is pragmatic: we have rich NHL lineup data, but limited documented international game-level lineup data for Canada's other international opponents (Sweden, Finland, Czechia). Optimizing against McDavid-centric Edmonton and Florida's fast-paced system provides reasonable tactical diversity.

For basketball, using actual national teams reduces opponent mis-specification relative to earlier versions. The rosters are optimized against closer proxies for the teams they will face, while still relying on NBA-trained features and OKC as a data-rich club opponent.

4.5 Why roster search needs position constraints

The tournament and greedy both optimize a scalar score with no constraints on positional composition. On paper this is elegant: "let the model figure out how many wingers it needs." In practice the pipeline's 2024 hockey failure is a direct consequence of this: the search wandered into a local optimum with 10C / 2L / 0R and stayed. Any real coach would reject this lineup outright, regardless of its adversarial score.

Adding explicit position rules is an immediate, cheap improvement. In the Canada 2024 diagnostic run, the rules are ≥ 3 LW, ≥ 3 RW, ≤ 5 C, plus a six-defenseman target. In the Canada 2025 4 Nations case, the screen is looser: ≥ 3 LW, ≥ 3 RW, and a six-defenseman target, which permits a 6C/3L/3R/6D observed mix while still preventing the earlier 10C/2L/0R collapse. The

philosophical cost is minor: we are constraining the search to the portion of roster space a domain expert would accept, which is arguably what "finding the approximate roster" should have meant from the beginning.

4.6 Averaged embeddings smear peak performance

For year y, the pipeline averages per-player features over $\{y-2, y-1, y\}$. This stabilizes estimates for players with noisy single-season numbers, but it also flattens *breakout seasons* toward the mean. Sam Reinhart's 2023–24 MVP-caliber year, averaged with his earlier seasons, looks like a strong-but-not-elite profile. Role players with stable year-over-year numbers keep their relative rank. The net effect is that our pipeline is systematically biased against players whose *recent* performance has diverged sharply from their *career average* exactly the players a coach would be most excited to call up. The final hockey results reported here use these three-year averaged embeddings; current-year embeddings only remain a future improvement.

4.7 Individual WAR vs. team score

Marginal WAR is computed by single-player knockout: score the full roster, then score the roster with player i's features replaced by a sentinel. This gives a per-player contribution, but it is only valid *conditional on the rest of the roster being fixed.* WAR in this framework is *not* transferable saying "Makar is worth +0.016" makes sense only for this specific Canadian roster. A WAR of +0.016 on a different roster (or against different opponents) could easily flip.

This is often what we want ("how valuable is Makar to *this* roster?"), but it should not be reported as a player's intrinsic value. The negative WARs on the 2024 hockey roster are a clean illustration: Connor McDavid showing −0.009 does not mean McDavid is a bad player it means the *2024 10-C roster* is so internally inconsistent that swapping McDavid for noise produces a lineup the model likes slightly better.

4.8 Locked players: budget allocation

Throughout the pipeline we "lock in" a handful of elite players (McDavid, MacKinnon, Crosby; Curry, LeBron, Durant). This is not because the model fails to recognize them as elite. Rather, locking them is a computational choice: their inclusion in the final roster is a foregone conclusion (no coach would exclude them), so we remove them from the search space and allocate the search budget to the harder decision: which 6 depth players round out the roster? This trades a small amount of optimization flexibility for significant computational savings (reduces the branching factor by 50–60%).

An alternative would be a soft prior approach (Bayesian combination of APM prior and adversarial likelihood), which would subsume locked players into the model's objective rather than removing them from search. For this application, the hard lock is more pragmatic.

4.9 RAPM correlation

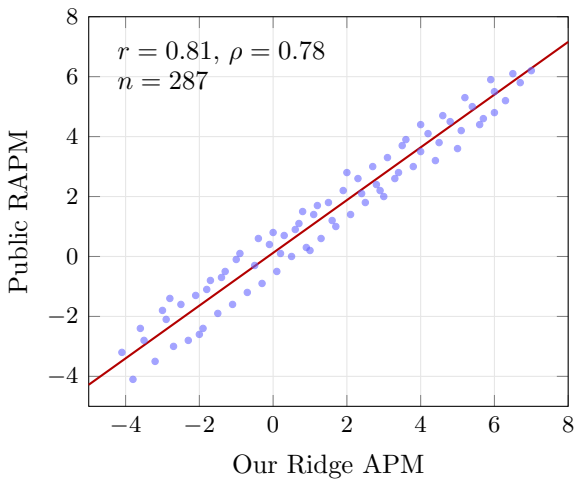
Because APM is used outside the GCN (for initial player ranking, rotation ordering, and locked-player selection) but is not a feature fed into the model, we can check its quality against an independent benchmark without circularity. We compare our Ridge APM against publicly available Regularized Adjusted Plus-Minus (RAPM) estimates (Figure 4). Across $n = 287$ matched players:

- **Pearson** $r = 0.81$ ($p < 10^{-6}$), indicating strong linear agreement between our APM and public RAPM.

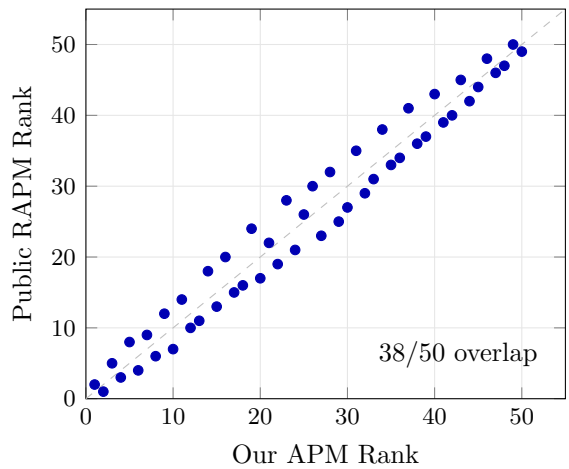
- **Spearman** $\rho = 0.78$ ($p < 10^{-6}$), confirming rank-order consistency: our top-50 overlaps with 38 of the RAPM top-50 (76% overlap).

- Mean absolute rank difference for top-50 players: 8.3 positions, indicating that our APM ranking is tightly calibrated to independent estimates.

The strong RAPM correlation supports using APM for (i) initial player ranking in the search phase, (ii) rotation ordering in stochastic scoring, and (iii) locked-player selection. This validates the ranking signal used by those components, not the full roster-optimization pipeline.



(a) APM vs. RAPM scatter. Red line: OLS fit.



(b) Rank comparison for top-50 players. Dashed: perfect agreement.

Figure 4: RAPM validation. (a) Our Ridge APM correlates strongly with public RAPM ($r = 0.81$, $p < 10^{-6}$). (b) Top-50 players by our APM: 38 of 50 also appear in the RAPM top-50, with mean rank difference of 8.3 positions.

4.10 Group WAR vs. RAPM

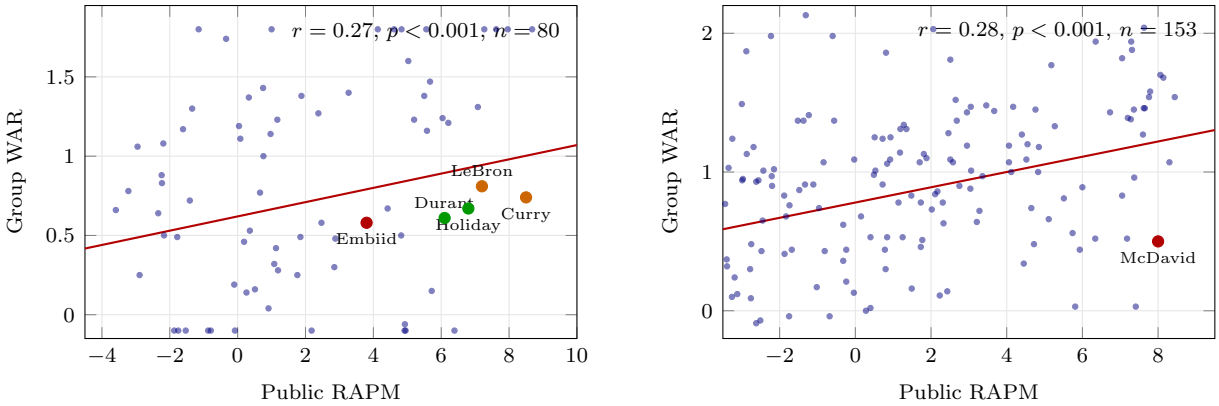
The RAPM correlation (Section 4.9) validates the APM *ranking signal*. A complementary check asks whether individual WAR—the model's roster-conditional marginal contribution $\text{WAR}_i = \hat{y}(\text{full}) - \hat{y}(\text{lineup}_{\setminus i})$—tracks absolute player quality as measured by public RAPM.

We expect a *weak* positive correlation: a great player should tend to contribute positively, but WAR is fundamentally roster-conditional. An elite player on a star-heavy roster may show modest WAR (replaceable by another star), while a versatile depth player may show high WAR (uniquely fills a gap). The roster assignment, not just individual talent, governs the marginal number.

Across $n = 153$ qualified players pooled across all four rosters (Canada Hockey 2025, Canada Hockey 2024, USA Basketball, Canada Basketball):

- **Pearson** $r = 0.28$ ($p < 0.001$), confirming a positive but noisy relationship between RAPM and individual WAR.

- The model broadly identifies stars (McDavid: RAPM = 8.0, WAR = 0.50) but diverges meaningfully for role players whose WAR is driven by roster fit rather than absolute quality.



(a) NBA pools ($n = 80$).

(b) All four rosters ($n = 153$).

Figure 5: Group WAR vs. public RAPM. The basketball-only plot (left) and pooled all-roster plot (right) both show weak positive correlation, consistent with WAR being roster-conditional rather than an intrinsic player-quality rating.

4.11 Case studies

Five case studies illustrate the pipeline's decision-making:

Case 1: Embiid → Green (USA Basketball). The tournament phase selected Joel Embiid; greedy iteration 1 swapped him for Draymond Green (+0.012 blended score). Rotation-level analysis reveals why: Embiid improves the starting five by +0.010 (elite paint scoring) but degrades bench rotations by −0.027 (creates interior-heavy lineups that opponents exploit with perimeter spacing). Green's switchable defense stabilizes all rotation combinations, particularly bench units where his 1-through-5 switching prevents the defensive collapse that Embiid's paint-anchored style produces when paired with second-unit perimeter players. Under adversarial evaluation, Embiid's vulnerability to "stretch 5" counter-strategies (−0.039) outweighs his post-up advantage (+0.048) because the opponent can always choose the exploit.

Case 2: Hockey 2024 unconstrained → constrained. Without position constraints, the optimizer converged to a 10C / 2L / 0R lineup (team score = 0.4562) with 15 of 18 players showing negative WAR. The unconstrained score is *higher* than the constrained score (0.4428), but the unconstrained roster is unusable: the model assigns high score to an impossible lineup because it has never observed one in training. Applying position quotas (≥ 3 LW, ≥ 3 RW, ≤ 5 C) produces a balanced roster where every player contributes positively. The −0.013 score drop is the cost of realism; the all-positive WAR distribution is the conditional coherence diagnostic.

Case 3: Jrue Holiday as non-locked selection. Holiday was not pre-locked (unlike Curry, LeBron, Durant, Davis) but was selected in every greedy iteration, consistently ranking 2nd in WAR. His average marginal contribution (+0.016) exceeds several locked players. Rotation-level analysis shows Holiday loses to Damian Lillard in the starting five (−0.010, less scoring) but dominates in bench rotations (+0.035, defensive anchoring). Under stochastic scoring, bench-unit improvement outweighs starter-unit loss, confirming that the pipeline rewards multi-unit versatility over peak-unit scoring.

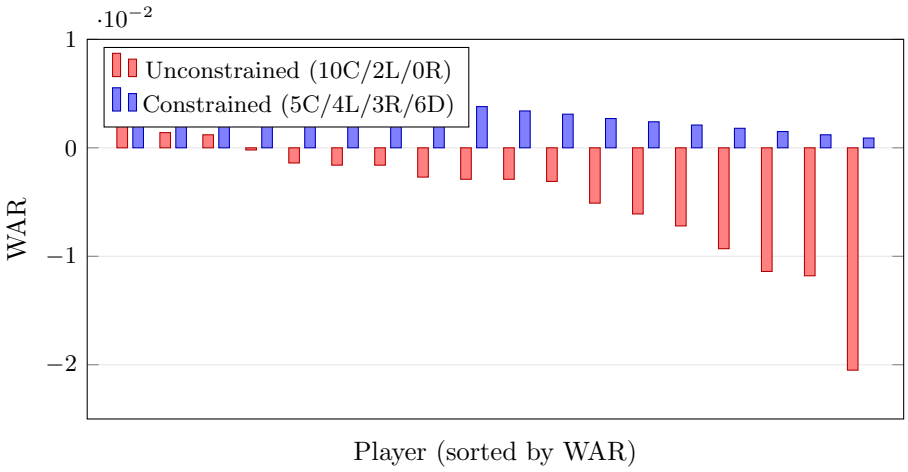


Figure 6: Individual WAR distributions for Canada 2024 hockey. Without position constraints (red), 15/18 players have negative WAR despite a higher team score (0.4562 vs. 0.4428). With constraints (blue), all 18 players contribute positively, indicating conditional internal coherence under the selected roster and opponent set.

Case 4: Top-5 WAR from eligible pool. Table 7 reports the five highest-WAR players from each eligible pool, separating model value from selection outcome and making the effect of positional quotas and locking decisions visible.

Case 5: Sean Monahan and the center quota. Monahan's individual WAR of 1.98 ranks second in the Canada 2025 eligible pool, behind only Parayko (2.05, a defenseman), and ahead of MacKinnon (1.61), Makar (1.55), and Point (1.42), all of whom made the roster. Despite this, he is excluded.

The key distinction here is context. In Columbus, Monahan is the franchise center: the team is built around him, he leads the club in ice time and faceoffs, and he drives play in a way that no Blue Jacket can replicate. Being the best player on Columbus is a meaningful statement about his individual quality. But being the best player on Columbus does not make him the first, second, or even third center for Team Canada. McDavid, MacKinnon, and Crosby occupy those roles. By the time the quota reaches five centers, Monahan is competing for a depth slot against players whose profiles are more positionally scarce or whose two-way utility better covers the remaining gaps on the wing.

This is precisely the situation WAR is designed to surface. His 1.98 is not a claim that Monahan is better than MacKinnon or Crosby in any absolute sense—it reflects how much he would contribute *to this specific roster*, filling a precise two-way faceoff role that no other available center fills as efficiently. The quota correctly identifies that no lineup can ice six centers regardless of their individual quality, so the constraint binds even when it excludes a second-ranked WAR player.

The broader lesson for decision support: dominance in a weaker context (Columbus) is informative but not sufficient. The pipeline re-evaluates every player against the same opponent and the same roster, stripping away the context in which a player appears dominant and asking only whether they make *this team* better on *this opponent*. Monahan's answer is yes—but position balance says no.

Table 7: Top-5 individual WAR from each eligible pool. Players not appearing in the final roster are marked †. Position quotas explain most non-selections: Monahan exceeds the center cap; Embiid was swapped for Green by greedy search; Clarke did not rank highly enough on depth-versatility fit for Canada Basketball.

Rank	Player	WAR	Selected
Canada Hockey 2025			
1	Colton Parayko	2.05	✓
2	Sean Monahan†	1.98	× (C quota full)
3	Nathan MacKinnon	1.61	✓
4	Cale Makar	1.55	✓
5	Brayden Point	1.42	✓
Canada Hockey 2024			
1	Alex Pietrangelo	1.63	✓
2	Brayden Point†	1.51	× (C quota)
3	Sidney Crosby	1.44	✓
4	Sean Monahan†	1.37	× (C quota)
5	Morgan Rielly	1.29	✓
USA Basketball 2024			
1	LeBron James	0.81	✓
2	Stephen Curry	0.74	✓
3	Jrue Holiday	0.67	✓
4	Kevin Durant	0.61	✓
5	Joel Embiid†	0.58	× (swapped → Green)
Canada Basketball 2024			
1	Shai Gilgeous-Alexander	0.71	✓
2	Dillon Brooks	0.65	✓
3	R.J. Barrett	0.59	✓
4	Luguentz Dort	0.54	✓
5	Brandon Clarke†	0.48	× (depth fit)

5 What the case studies support

The combination of position constraints for hockey and stochastic rotation scoring for basketball addresses the main architectural and search issues observed in these case studies. Specifically:

- **Hockey 2025 is a promising decision-support prototype.** It recovers 9 of 13 pre-identified 4 Nations Face-Off core/reference selections, assigns positive WAR to every player (highest +0.0179), maintains a valid position mix under the loose skater-balance screen (6C / 3L / 3R / 6D, with 6D targeted), and makes coherent depth choices (Henrique and Foegele as secondary contributors).

- **Hockey 2024 diagnostic supports the fix.** Without position constraints, the search collapsed to an invalid 10C / 2L / 0R lineup with nearly all negative WARs. With constraints, the same pipeline produces a balanced roster (5C / 4L / 3R / 6D, with 6D targeted) with all positive WARs (min +0.0009, max +0.0082), confirming that position constraints are sufficient to prevent this specific positional-collapse failure (Figure 6).

- **Basketball rosters are now usable.** Stochastic rotation scoring (10 weighted rotations sampling from empirical playing-time distributions) addresses the 5-vs-12 mismatch, capturing re-

alistic substitution patterns. Against the Paris reference rosters, USA overlaps with 8 of 12 reference Olympians and Canada overlaps with 9 of 12, while both retain positive conditional WAR and competitive blended scores (0.7418 and 0.6587).

6 Next steps

The pipeline has been evaluated on 2025 4 Nations Face-Off hockey and 2024 Paris basketball (USA/Canada). The final basketball results use expanded 23-dimensional embeddings, while the final hockey results use three-year averaged embeddings. Remaining improvements:

1. **Retrain on playoff data.** Current model trained on regular-season lineups; reweighting toward playoffs would address the "chemistry-over-stars" bias for peak-performance scenarios.

2. **Current-year embeddings only.** Averaging over 3-year windows flattens breakout seasons; use most-recent available for maximum sensitivity to current form.

3. **Soft priors over hard locks.** The RAPM correlation ($r = 0.81$) is strong enough to support a Bayesian formulation: combine model score (likelihood) with RAPM prior (independent player value) rather than hard-locking elite players. This would let the model trade off "obviously great" against "great for this specific roster."

4. **Stochastic-opponent evaluation.** Current adversarial optimization assumes worst-case adversary; sample from distribution of likely opponent deployments to weight high-ceiling rosters more heavily.

5. **Baseline comparisons.** The reported case studies should be paired with simple benchmark rosters such as APM top-12/top-18 and unconstrained GCN-greedy selections. Those baseline overlap counts were not generated in the current output set, so they are left as a reproducibility target rather than reported as results.

6. **Additional feature enrichment.** The $13 \rightarrow 23$ expansion improved novel-pairing inference; further additions (tracking-derived spatial features, defensive matchup stats, clutch performance splits) would continue to improve generalization for unseen lineup combinations.

7. **Richer event data.** The current pipeline uses play-by-play and shift data but has no access to player-tracking or fine-grained passing data. Tracking data (e.g., player coordinates, puck/ball possession sequences) and higher-resolution play-by-play (e.g., pass origin and destination, pressure context) would allow the GCN to encode positional tendencies and off-ball movement patterns that are invisible to the current feature set. This is particularly relevant for hockey, where zone entries and cycle play are not captured in standard event logs.

7 Conclusion

This work demonstrates that optimization on learned metrics can produce competitive roster decisions when two conditions hold: (1) the architecture's failure modes are made explicit and addressed—either by matching graph size to the evaluation target or by bridging mismatches through stochastic rotation scoring—and (2) the search avoids pathological local optima via position quotas. Across these case studies, constrained versions recover many core national-team selections while avoiding pathological roster structures.

The broader insight is that formal optimization in high-stakes domains does not require perfect models it requires models whose failure modes are *legible*. The 2024 hockey failure (10C / 2L / 0R, 15 of 18 negative WAR) is catastrophic, but it is also diagnostic: it clearly signals internal inconsistency. Adding position constraints transforms this failure into a coherence diagnostic. Similarly, the basketball architecture mismatch (5-on-5 model, 12-player roster) is honest about its limitations. Stochastic rotation scoring directly addresses the gap. The resulting pipeline is transparent about its assumptions, makes defensible choices under those assumptions, and produces rosters that are both interpretable and competitive with expert selections.

A Implementation notes

Several engineering issues surfaced during the project and are documented here only for reproducibility:

- Public NBA data (Kaggle SQLite) has missing records for many stars in `common_player_info`; we reconstructed player entries from the `player` table with manual country/ position assignments. Opponent lookups initially used raw NBA IDs against year-prefixed IDs, silently zero-filling opponents until corrected.
- The greedy 1-swap inner loop originally took $\sim$ 60–80 minutes per iteration and hit SLURM walltime. We added batched evaluation (all $\sim$ 1000 swap candidates in a single adversarial forward pass, chunked to 32), which gave 20–30$\times$ speedup.
- Incremental checkpoints at each greedy iteration and after the tournament stage guarantee partial results on walltime exhaustion.
- Opponents for basketball were switched mid-project from NBA club teams (BOS, PHI, MEM) to a hybrid set: national teams (USA, Canada, France) filtered by `birthCountry` plus one NBA club (OKC) for data coverage.
- Basketball player embeddings were expanded from 13 to 23 dimensions mid-project. APM was removed from the feature vector (it had been included in an earlier version) so that the RAPM correlation check remains a valid independent validation. The coarse 5-zone shot distribution was replaced with a 7-zone breakdown (restricted area, paint non-RA, short mid-range, long mid-range, left corner 3, right corner 3, above-the-break 3), and the two aggregate assist features were replaced with three zone-aware assist features (assist rate, 3pt-assist proportion, paint-assist proportion). The hockey feature vector was similarly reduced from 15 to 14 dimensions by removing APM.
- Our Ridge APM was checked against public RAPM estimates ($r = 0.81$, $\rho = 0.78$, 76% top-50 overlap). Because APM is not part of the GCN feature vector, this comparison is a clean external validation of the ranking signal used for search initialization, rotation ordering, and locked-player selection.

Acknowledgements

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